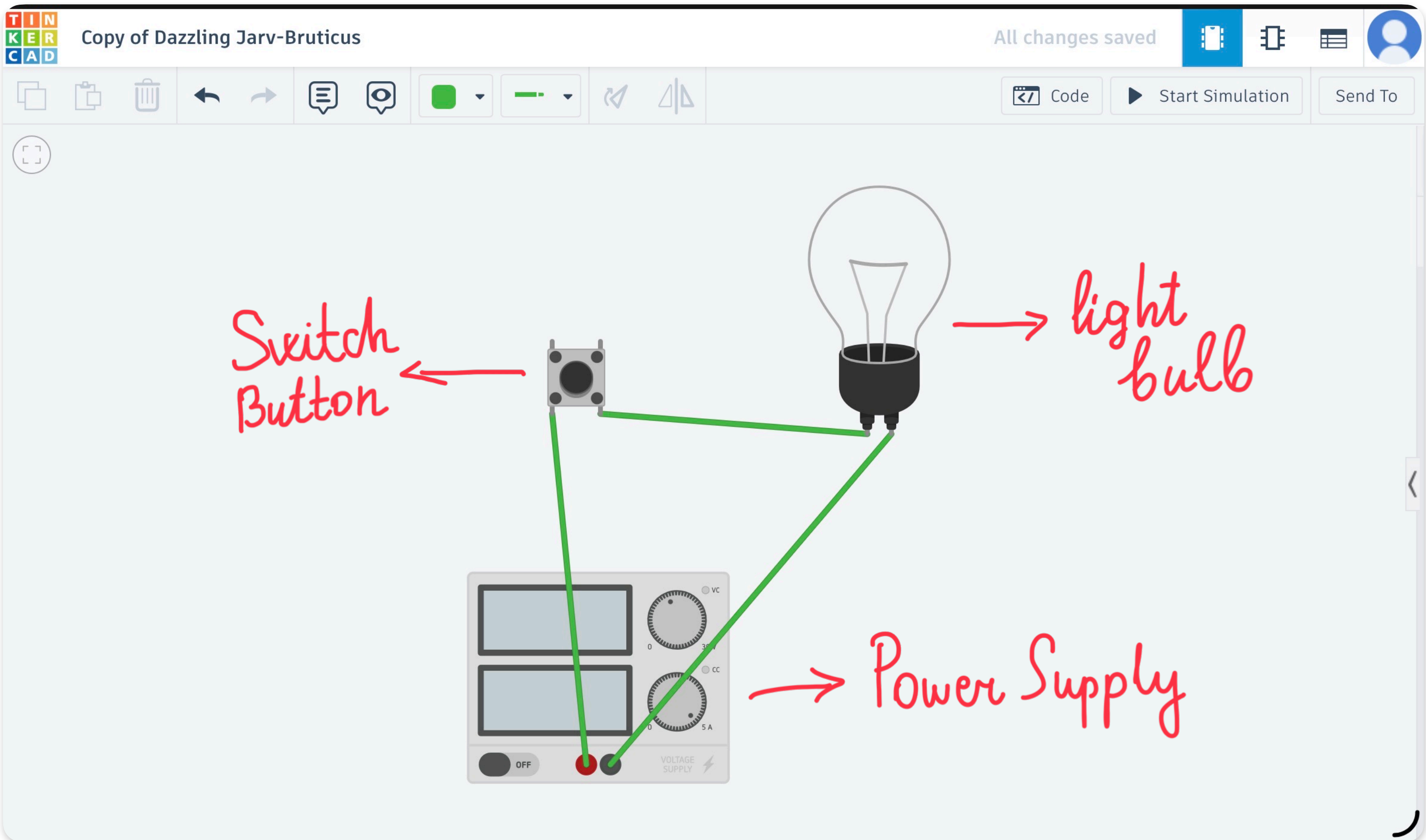
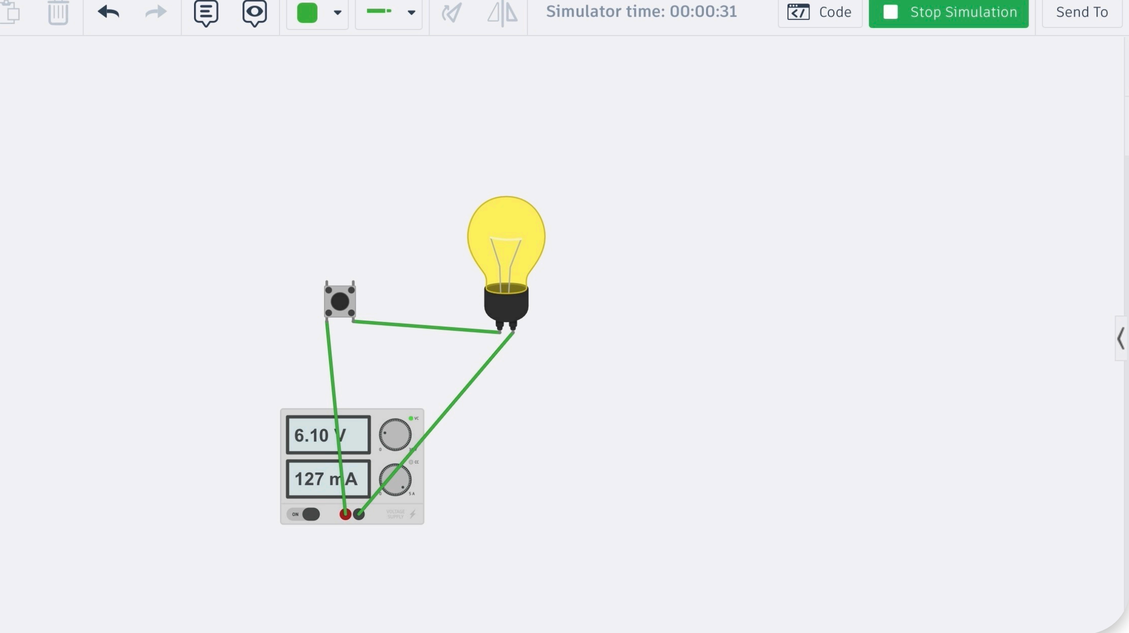
In this circuit we use 3 components.
Our goal is to find the threshold power value of the circuit to avoid from light bulb blow up.

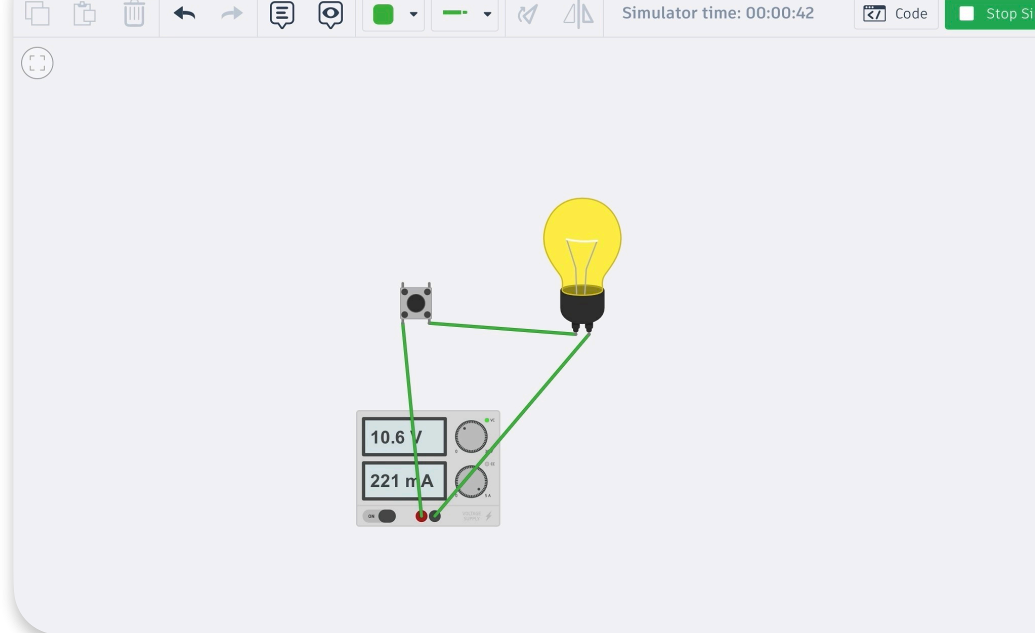


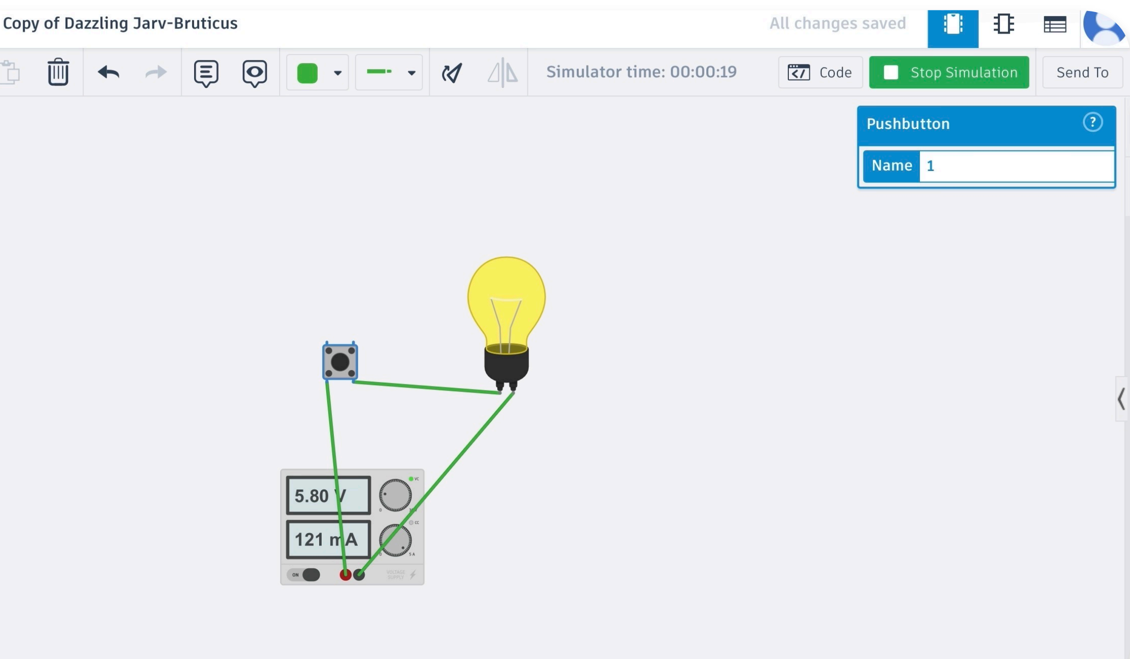
Formulas we use in this experiment are:

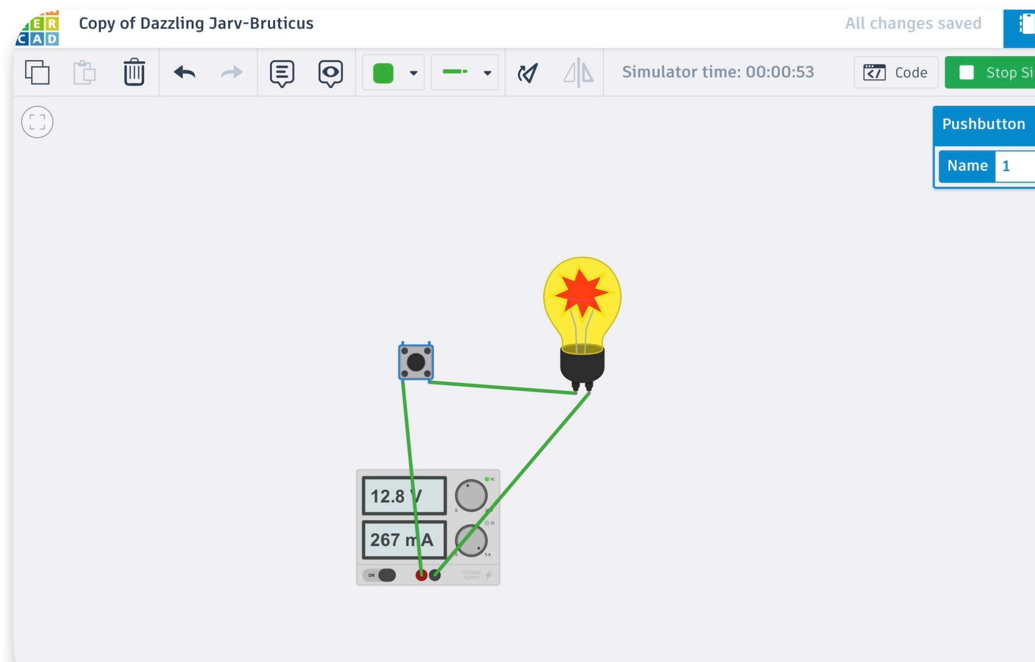
$$\underbrace{P}_{\text{Power}} = \underbrace{V}_{\text{Voltage}} \cdot \underbrace{I}_{\text{Current}}$$

$$\underbrace{R}_{\text{Resistor}} = \frac{V}{I}$$


Simulator time: 00:00:31
Code
Stop Simulation
Send To
6.10 V
127 mA


Simulator time: 00:00:42
Code
10.6 V
221 mA


Copy of Dazzling Jarv-Bruticus
All changes saved
Simulator time: 00:00:19
Code
Stop Simulation
Send To
Pushbutton
Name 1
5.80 V
121 mA


Copy of Dazzling Jarv-Bruticus
All changes saved
Simulator time: 00:00:53
Code
Pushbutton
Name 1
12.8 V
267 mA

We applied different voltage values to the power supply and gradually increased the voltage. When the voltage reached 12.8 V, the light bulb blew up (burned out). Based on our investigation and observations, we concluded that if the voltage exceeds 12 V and the power becomes greater than 3 W, the light bulb fails and an explosion (burnout) occurs.

Light bulb working/blow up schema

V	I	R	P	Light bulb
5.8 V	121mA	47.93 Om	0.7 W	Works
6.1 V	127mA	48.03 Om	0.77 W	Works
10.6 V	221mA	47.96 Om	2.34 W	Works
12.8 V	267mA	47.94 Om	3.41 W	Blow up

1. Maximum Limits (No Blow-Up)

- **Voltage:** 12V DC (Standard 9V battery or a 12V setting on the Power Supply component is ideal).
- **Current:** Keep the amperage below **250 mA (0.25A)**.
- **Power Calculation:** Using $P = V \times I$, a safe maximum power is $12\text{V} \times 0.25\text{A} = \mathbf{3}\ \text{Watts}$. Course Hero +2

What we learned:

- **How to use the formulas $P = I \times V$ and $R = V / I$ to calculate power, current, voltage, and resistance in a circuit.**
- **The operating limits of a light bulb and the conditions that cause it to burn out (explode).**
- **That increasing voltage leads to higher power, and when it exceeds the safe limit (around 12 V and 3 W), the bulb fails.**
- **The working principle of a power supply and how adjusting the voltage affects current and power in the circuit.**
- **The importance of correct electrical calculations to prevent damage to circuit components.**